



Amazon Interview Experience

Amazon is an American multinational technology company focusing on e-commerce, cloud computing, online advertising, digital streaming, and artificial intelligence. It is considered one of the Big Five American technology companies, alongside Alphabet (parent company of Google), Apple, Meta (parent company of Facebook), and Microsoft. It is a product-based company and one of the famous MAANG companies.

<i>Role Offered</i>	-	<i>SDE – 6 months Intern (placement based on performance)</i>
<i>Stipend</i>	-	<i>1,10,000 + 400 USD / Month</i>
<i>Number of Rounds</i>	-	<i>2 (Both Online).</i>

The Assessment pattern may change for every year. This year it was only coding round and personality questions whereas they would also add debugging, Aptitude as well.

This year the selection was based on two rounds: one online coding round and an interview.

Round 1: (ONLINE ASSESSMENT)

It consists of two phases:

- *Coding.*
- *Personality questions.*

The online assessment consists of two coding questions, which have to be done in 70 minutes. This round was held in **HackerRank** platform.

Coding Question 1:

This question was based on regex pattern.

A regex pattern consists of lowercase English letters and some wildcard characters (“?”) with any English character. Two patterns are called intersecting if the wildcard characters in the respective string can be changed in a way to form the same string from both patterns.

For Example: pattern $p = \text{"?cc?"}$ and $q = \text{"dc??"}$ intersect as we can form $s = \text{"dccx"}$ from both p and q .

Given an array pattern of N regex strings, find the minimum number of “?” characters possible in a pattern that intersects with all the patterns.

Sample Input:

Given n = 2, m = 10, patterns = [“ha??rrank”, “?a?ke?bank”]

Sample output:

So here the minimum number of wildcard characters that we need to generalize this array is 1.

Coding Question 2:

Problem Statement

You are given:

- An integer array of parcels, where each element represents the number of parcels each agent currently has.
- An integer extra_parcels representing the total number of extra parcels that need to be distributed among the agents.
Distribute the extra parcels such that the maximum number of parcels any agent has after distribution is minimized. You need to return this minimized maximum number.

Example:

Input:

parcels = [7, 5, 1, 9, 1]

extra_parcels = 25

Output:

10

Explanation:

One possible way to distribute the parcels is:

- Agent 1: 7 parcels + 3 extra = 10 parcels
- Agent 2: 5 parcels + 4 extra = 9 parcels
- Agent 3: 1 parcel + 9 extra = 10 parcels

- Agent 4: 9 parcels + 1 extra = 10 parcels
- Agent 5: 1 parcel + 8 extra = 9 parcels

There could be other distributions, but the minimum possible maximum number of parcels is 10.

After the coding round you will be asked to answer some personality-based questions there will be no marking for this round and there is no time constraints too. In this round the same questions may be repeated so ensure that you are thoroughly reading the questions and answer the same for the repeated questions. This will evaluate your personality and how you are thinking to solve some scenarios.

Based on this online assessment, around 85 people were shortlisted for the interview.

Round 2: (TECHNICAL INTERVIEW)

Mode: ONLINE.

Before the interview, you will be allotted a slot (timing for the interview), and the link for the meet and **LiveCode** link will be shared via email once you are registered on the Amazon careers.

The interview was conducted in **Amazon's Chime** platform.

For me there was two panels.

The interview started with the introduction of the interviewers and then I was asked to introduce myself. Suggestion: (Keep your introduction short and crisp since the interview session will be 1 hour and the interviewer will have more time on questions.) And also, there will be no questions based on the resume and your soft skills ensure that you add those points when you introduce yourself).

Then you would be asked to share the screen and open the LiveCode session link you have received on your email.

The interviewer asked how strong I am in Heap data structure and asked me some questions on HEAP.

1. What is Heap?
2. Types of Heaps?
3. How heap maintain its structure?
4. How heapify will happen if an element is deleted?
5. Time complexity for Heapify.

6. How you derive the time complexity for Heapify?
7. Given one tree and asked whether it is a heap or not?

After answering all questions then he started with some questions on Tree

1. What is a binary tree?
2. Difference between a complete and full binary tree?
3. Asked about the time complexity for the traversal?
4. Given a number of nodes find the height for both complete and full binary tree?
5. Asked about how you derive the formula for finding the height of the tree.

I answered all questions, and the interviewer was satisfied with my answers.

Then he moved on to the coding question.

Question Asked:

Step-by-Step Directions from a Binary Tree Node to Another

You are given the root of a **binary tree** with n nodes. Each node is uniquely assigned a value from 1 to n . You are also given an integer `startValue` representing the value of the start node s and a different integer `destValue` representing the value of the destination node t .

Find the **shortest path** starting from node s and ending at node t . Generate step-by-step directions of such path as a string consisting of only the **uppercase** letters 'L', 'R', and 'U'. Each letter indicates a specific direction:

Firstly, I was asked to explain my intuition and to code the solution.

Since there is no brute force approach for this problem, I directly gave the solution and coded the same.

After coding the solution for this question, the interviewer asked for some test cases since the I was coded for all edge cases in the first attempt, I can answer the questions very easily and the interviewer was satisfied with my solution.

Then asked for the time and space complexity.

Then some questions on the basics of programming from my code since I wrote the code in CPP, he asked a few questions like

1. Why do you pass the argument as a reference?
2. Why not pointer, why reference?
3. Since I used some built-in functions, I was asked how this function will work and explain the pseudocode for the same.

The interview ended with “is there any question for me”. This is the first time in an interview I asked no question at the end I don’t know why? But the interviewer seemed happy.

But make sure you ask some questions at the end.

RESULT:

There were fifteen people offered six-month internships out of eighty-five, and I was one of them.

Suggestions:

- Amazon’s interview is different from all interviews they don’t ask any questions from your resume. They mainly focus on your thought process for solving the problem. And make the interview like a discussion explain all the things that you are thinking of the question. Use the whiteboard that was shared with you to make the explanation clear. You can use paper and pen but yet make sure you use that whiteboard.
- Please go through Amazon’s leadership principles before the interview. Not only for Amazon, but for all the companies, do the research before the interview.

General Suggestion:

- Start your coding journey earliest and practice problems on a daily basis. Prefer quality over quantity numbers doesn’t matter.
- Don’t feel bad if the result doesn’t go on your way. Be confident and try until you succeed.
- Get strong fundamentals in DSA, OOPS, DBMS, OS, and CN and brush up some system design concepts.
- After the online assessment you may not have enough time to prepare for core concepts so be ready beforehand.

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